

TOPICS IN MATHEMATICAL SCIENCE

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INTRODUCTION TO KOSZUL DUALITY

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Convention

Throughout the course, $\mathbb{k}$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\mathbb{k}$ -vector spaces.

1 Rings and modules

1.1 Semisimple algebras

Definition 1.1. A $\mathbb{k}$ -algebra is a $\mathbb{k}$ -vector space with an associative unital multiplication that is *bilinear*, meaning linear in each variable.

A right A -module is understood to be a $\mathbb{k}$ -vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$.

Example 1.2. (1) The matrix ring $\text{Mat}_n(\mathbb{k})$ is a $\mathbb{k}$ -algebra. Every row space (the space of $1 \times n$ -matrices) is a $\text{Mat}_n(\mathbb{k})$ -module.

(2) Let $\mathbb{k}A_n$ be the *upper triangular matrix ring*, which is also a $\mathbb{k}$ -algebra.

$$\mathbb{k}A_n = \begin{pmatrix} \mathbb{k} & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & \mathbb{k} & \cdots & \vdots \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \mathbb{k} \end{pmatrix}$$

For those familiar with path algebras, $\mathbb{k}A_n$ is the path algebra of a *linearly oriented A_n quiver*, which explains the notation. We use matrix calculations, so no knowledge of quivers is required.

Definition 1.3. A module L is *simple* if $L \neq 0$ and its only submodules are 0 and L . A module is *semisimple* if it is a direct sum of simple modules. An algebra B is *semisimple* if it is semisimple as a module B_B over itself (this does not depend on which side you work on). The *Jacobson radical* $\text{rad } B$ is the intersection of all maximal right ideals of B .

We use the following standard facts without proof. Recall that a *division $\mathbb{k}$ -algebra* is a non-zero $\mathbb{k}$ -algebra whose non-zero elements are invertible.

Theorem 1.4 (Artin–Wedderburn). For a non-zero finite-dimensional $\mathbb{k}$ -algebra B ,

$$B \text{ is semisimple} \iff \text{rad } B = 0 \iff B \cong \prod_{i=1}^r \text{Mat}_{n_i}(D_i),$$

where $r, n_i \geq 1$ and each D_i is a finite-dimensional division $\mathbb{k}$ -algebra. If $\mathbb{k}$ is algebraically closed, then $D_i = \mathbb{k}$ for every i . For semisimple B , every finite-dimensional left or right B -module is semisimple, and every submodule has a complement.

Example 1.5. Let e_{ij} be the unit matrix with 1 at the (i, j) -entry and zero everywhere else. Then we have $\text{rad}(\mathbb{k}A_2) = \mathbb{k}e_{12} \neq 0$, and so $\mathbb{k}A_2$ is not semisimple.

1.2 Graded algebras

Definition 1.6. Let G be a group with identity e . A G -graded $\mathbb{k}$ -algebra is a $\mathbb{k}$ -algebra A with $1_A \in A_e$ and a decomposition

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is *homogeneous of degree g* .

Convention. Unless stated otherwise, a graded algebra is $\mathbb{Z}$ -graded and concentrated in non-negative degrees.

Note that the decomposition is into a direct sum, so every element of a graded algebra A can always be written as a *finite* sum of homogeneous elements.

Example 1.7. For the upper triangular matrix ring $\mathbb{k}A_n$ of Example 1.2, the decomposition

$$(\mathbb{k}A_n)_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}$$

makes $\mathbb{k}A_n$ a graded algebra.

Example 1.8. The algebra $\mathbb{k}A_3$ also has a $\mathbb{Z}^{\oplus 2}$ -grading with

$$\deg e_{ii} = (0, 0), \quad \deg e_{12} = (1, 0), \quad \deg e_{23} = (0, 1), \quad \deg e_{13} = (1, 1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$, $(p, q) \mapsto p + q$, gives the grading of Example 1.7. Alternatively, setting $\deg e_{ij} = 2(j - i)$ gives a $\mathbb{Z}$ -grading with $A_1 = 0$ but $A_2 \neq 0$.

Definition 1.9. A *homomorphism of graded algebras* $f : A \rightarrow B$ is a $\mathbb{k}$ -algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.

Definition 1.10. A graded algebra A is *locally finite* if $\dim_{\mathbb{k}} A_i < \infty$ for every $i \in \mathbb{Z}$.

1.3 Graded modules

Definition 1.11. Let $A = \bigoplus_{g \in G} A_g$ be a G -graded algebra. A G -graded right A -module is a right A -module M with a direct sum decomposition

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called *homogeneous of degree g* .

A G -graded A -module $M = \bigoplus_{g \in G} M_g$ is *locally finite-dimensional* if $\dim_{\mathbb{k}} M_g < \infty$ for all $g \in G$.

Unless specified otherwise, graded modules are $\mathbb{Z}$ -graded; their degrees need not be non-negative. Write $\text{Mod } A$ for the category of right A -modules, $\text{GrMod } A$ for graded right A -modules with degree 0 homomorphisms, and $\text{grmod } A$ for its full subcategory of locally finite-dimensional modules.

The *opposite algebra* A^{op} has the same vector space as A , with multiplication $a^{\text{op}} b^{\text{op}} = (ba)^{\text{op}}$. A left A -module is a right A^{op} -module by $m \cdot a^{\text{op}} = am$. For a $\mathbb{Z}$ -graded algebra, $(A^{\text{op}})_i = A_i$.

Definition 1.12. An *A-bimodule* is a vector space M with $\mathbb{k}$ -bilinear left and right A -module actions satisfying $(am)b = a(mb)$ for all $a, b \in A$ and $m \in M$. It is graded if $M = \bigoplus_{j \in \mathbb{Z}} M_j$ and $A_i M_j A_k \subset M_{i+j+k}$ for all $i, j, k \in \mathbb{Z}$.

The algebra A is a graded A -bimodule by multiplication, and each A_i is an A_0 -bimodule. The left and right scalar actions agree: $(c1_A)m = cm = m(c1_A)$ for $c \in \mathbb{k}$.

Example 1.13. The decomposition $A = \bigoplus_{i \in \mathbb{Z}} A_i$ makes the right regular module A_A a graded module.

Definition 1.14. Let M be a graded A -module and let $d \in \mathbb{Z}$. The *shift* $M(d)$ is the graded A -module whose underlying A -module is M , but whose grading is

$$M(d)_j = M_{j+d}.$$

By Definition 1.14, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d .

Example 1.15. For $A = \mathbb{k}A_3$ with the grading of Example 1.7, the module $P_1 := e_{11}A$ consists of matrices supported on the first row. Its basis e_{11}, e_{12}, e_{13} has degrees 0, 1, 2, respectively; in $P_1(-2)$, the same basis has degrees 2, 3, 4.

Definition 1.16. Let M, N be graded A -modules. A homomorphism $f : M \rightarrow N$ is *homogeneous of degree d* if $f(M_j) \subset N_{j+d}$ for all $j \in \mathbb{Z}$. A *homomorphism of graded modules* means a homogeneous A -module homomorphism of degree 0, also said to *preserve degrees*.

Notation. For graded right A -modules M, N , $\text{Hom}_A(M, N)$ denotes all right A -module homomorphisms, without a grading condition. We use

$$\begin{aligned} \text{Hom}_{\text{GrMod } A}(M, N) &= \{f \in \text{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}, \\ \underline{\text{Hom}}_A(M, N) &= \bigoplus_{d \in \mathbb{Z}} \text{Hom}_{\text{GrMod } A}(M, N(d)). \end{aligned} \quad (1.1)$$

The degree d component of $\underline{\text{Hom}}_A(M, N)$ consists of homomorphisms $M \rightarrow N$ of degree d , since $N(d)_j = N_{j+d}$. For left modules we use $\text{Hom}_{A^{\text{op}}}$ and $\text{Hom}_{\text{GrMod } A^{\text{op}}}$.

There is a natural inclusion $\underline{\text{Hom}}_A(M, N) \subset \text{Hom}_A(M, N)$: an element on the left is a finite sum of homogeneous homomorphisms. Equality holds when M is finitely generated, but need not hold for arbitrary M .

Example 1.17. For $A = \mathbb{k}A_3$, left multiplication by e_{12} gives

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (u, v \in \mathbb{k}). \end{aligned}$$

The u and v entries have degrees 0, 1 in the source and 1, 2 in the target. Thus the map has degree 1; after replacing the source by $e_{22}A(-1)$, it belongs to $\text{Hom}_{\text{GrMod } A}(e_{22}A(-1), e_{11}A)$. Right A -linearity follows from associativity of matrix multiplication.

1.4 Homogeneous ideals and quotients

Definition 1.18. Let $A = \bigoplus_{i \in \mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is *homogeneous* if

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I .

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i / (I \cap A_i)$$

is again a graded algebra.

Example 1.19. In $A = \mathbb{k}A_3$ with the grading of Example 1.7, the two-sided ideal $I = (e_{13}) = \mathbb{k}e_{13}$ is homogeneous. Writing $\bar{e}_{ij}$ for the image of e_{ij} ,

$$(A/I)_0 \cong \mathbb{k}^{\oplus 3}, \quad (A/I)_1 = \mathbb{k}\bar{e}_{12} \oplus \mathbb{k}\bar{e}_{23}, \quad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\bar{e}_{12}\bar{e}_{23} = 0$.

Example 1.20. In $A = \mathbb{k}A_2$, the subspace $I = \mathbb{k}(e_{22} + e_{12})$ is a right ideal: $Ie_{11} = Ie_{12} = 0$ and $Ie_{22} = I$. However, $e_{22}, e_{12} \notin I$, so I is not homogeneous. The right module A/I therefore does not inherit the quotient grading.

Definition 1.21. Let A be a graded algebra and set

$$A_{>0} := \bigoplus_{i>0} A_i.$$

The subspace $A_{>0}$ is a homogeneous two-sided ideal, also called the [augmentation ideal](#). We use the term [positive degree ideal](#).

The quotient map, also called the augmentation map,

$$\pi : A \longrightarrow A/A_{>0} \cong A_0, \quad \sum_{i \in \mathbb{Z}} a_i \mapsto a_0,$$

makes A_0 a graded A -bimodule concentrated in degree 0, with $a \cdot s \cdot b = \pi(a)s\pi(b)$ for $a, b \in A$ and $s \in A_0$.

1.5 Tensor products of vector spaces

Definition 1.22. For vector spaces U, V , the [tensor product](#) $U \otimes_{\mathbb{k}} V$ is equipped with a bilinear map $(u, v) \mapsto u \otimes v$ such that every bilinear map $b : U \times V \rightarrow W$ has a unique linear factorisation:

$$\begin{array}{ccc} U \times V & \xrightarrow{(u,v) \mapsto u \otimes v} & U \otimes_{\mathbb{k}} V \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

If $(u_i)_{i=1}^r$ and $(v_j)_{j=1}^s$ are bases, then $(u_i \otimes v_j)_{i,j}$ is a basis of $U \otimes_{\mathbb{k}} V$; hence $\dim_{\mathbb{k}}(U \otimes_{\mathbb{k}} V) = rs$. We use the canonical identifications $(U \otimes_{\mathbb{k}} V) \otimes_{\mathbb{k}} W \cong U \otimes_{\mathbb{k}} (V \otimes_{\mathbb{k}} W)$ and $\mathbb{k} \otimes_{\mathbb{k}} U \cong U$, and write $U^{\otimes 0} = \mathbb{k}$.

1.6 Graded $\mathbb{k}$ -linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space, we retain only finite sums of homogeneous functionals.

Definition 1.23. For a vector space V , its $\mathbb{k}$ -linear dual is $D(V) := \text{Hom}_{\mathbb{k}}(V, \mathbb{k})$. For a graded vector space $V = \bigoplus_{i \in \mathbb{Z}} V_i$, its *graded $\mathbb{k}$ -linear dual* is the graded vector space given by

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i \quad \text{with} \quad \mathbb{D}(V)_i = D(V_{-i}).$$

For general infinite-dimensional V , we have $\mathbb{D}(V) \subset D(V)$, whereas $D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i)$. These spaces coincide when V is finite-dimensional.

Linear duality reverses the side of a module action. For a right A -module M and a left A -module N , define

$$(af)(m) := f(ma), \quad (ga)(n) := g(an) \quad (a \in A, f \in D(M), g \in D(N)). \quad (1.2)$$

Here $m \in M$ and $n \in N$; the first formula makes $D(M)$ a left module and the second makes $D(N)$ a right module. For graded modules, the same formulas preserve the graded $\mathbb{k}$ -linear dual: if $a \in A_p$ and $f \in \mathbb{D}(M)_q$, then af vanishes outside M_{-p-q} and hence has degree $p + q$. There are no signs in these actions.

Example 1.24. Let $A = \mathbb{k}A_3$ and $P_1 = e_{11}A$ as in Example 1.15. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1 \leq j, \ell \leq 3$ form a basis of $\mathbb{D}(P_1)$, with

$$\deg \varepsilon_{1j} = 1 - j, \quad e_{12}\varepsilon_{12} = \varepsilon_{11}, \quad e_{23}\varepsilon_{13} = \varepsilon_{12}, \quad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.2). For comparison with an infinite-dimensional space, take $V = \bigoplus_{i \geq 0} \mathbb{k}v_i$ with $\deg v_i = i$. The functional $\lambda(v_i) = 1$ for every i belongs to $D(V)$ but not to $\mathbb{D}(V)$.

Lemma 1.25. $\mathbb{D}$ is a duality between $\text{grmod } A$ and $\text{grmod } A^{\text{op}}$: it reverses arrows and short exact sequences, and evaluation gives a graded module isomorphism $M \cong \mathbb{D}(\mathbb{D}(M))$ for every locally finite graded module M .

Proof A homomorphism $u : M \rightarrow N$ of degree 0 induces $\mathbb{D}(u) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$ by $f \mapsto f \circ u$. Dualising a short exact sequence in each degree is exact, since every linear functional on a subspace extends to the whole vector space. The evaluation map sends m to $(f \mapsto f(m))$. It is an isomorphism in each degree by finite-dimensional linear algebra, and (1.2) makes it a module homomorphism. $\square$

Left, right, and two-sided duals over S

Fix a finite-dimensional semisimple $\mathbb{k}$ -algebra S , concentrated in degree 0. Besides the $\mathbb{k}$ -linear dual, we need duals defined by left S -linear maps, right S -linear maps, and maps respecting both actions. Their definitions differ, even though the resulting spaces can be identified after a choice of form.

Definition 1.26. For a right S -module M and a left S -module N , define their *S -duals* by

$$\begin{aligned} M^* &:= \text{Hom}_S(M, S), & (sf)(m) &:= sf(m), \\ {}^*N &:= \text{Hom}_{S^{\text{op}}}(N, S), & (gs)(n) &:= g(n)s. \end{aligned}$$

Here $s \in S$, $m \in M$, $n \in N$, $f \in M^*$, and $g \in {}^*N$.

Thus M^* is a left S -module and *N is a right S -module. The position of the star records the side of the original action: $f(ms) = f(m)s$ in M^* , whereas $g(sn) = sg(n)$ in *N .

Definition 1.27. The *enveloping algebra* of S is $S^{\text{en}} := S \otimes_{\mathbb{k}} S^{\text{op}}$, with multiplication

$$(s \otimes t^{\text{op}})(s' \otimes (t')^{\text{op}}) = ss' \otimes (t't)^{\text{op}} \quad (s, t, s', t' \in S).$$

An S -bimodule is equivalently a right S^{en} -module via

$$m \cdot (s \otimes t^{\text{op}}) = tms \quad (s, t \in S, m \in M). \quad (1.3)$$

For example, $(m \cdot (s \otimes t^{\text{op}})) \cdot (s' \otimes (t')^{\text{op}}) = t'tmss' = m \cdot (ss' \otimes (t't)^{\text{op}})$. All three spaces $D(M)$, M^* , and *M inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \quad (sft)(m) = sf(tm), \quad (sgt)(m) = g(ms)t, \quad (1.4)$$

where $\lambda \in D(M)$, $f \in M^*$, $g \in {}^*M$, $s, t \in S$, and $m \in M$.

Definition 1.28. The *two-sided dual* of an S -bimodule M is

$$M^{*\text{en}} := \text{Hom}_{S^{\text{en}}}(M, S^{\text{en}}), \quad (sFt)(m) := (s \otimes t^{\text{op}})F(m) \quad (s, t \in S, F \in M^{*\text{en}}, m \in M).$$

In Definition 1.28, M has the right action (1.3), so $F(tms) = F(m)(s \otimes t^{\text{op}})$. The output is a left S^{en} -module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in S^{en} , not in S .

1.7 Comparing the duals

An S -dual consists of S -valued maps; the $\mathbb{k}$ -linear dual consists of scalar-valued maps. A suitable linear functional $S \rightarrow \mathbb{k}$ identifies these constructions.

Definition 1.29. A *symmetrising form* on a finite-dimensional $\mathbb{k}$ -algebra S is a linear map $\tau : S \rightarrow \mathbb{k}$ satisfying

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \quad (\tau(st) = 0 \text{ for all } t \in S) \implies s = 0.$$

An algebra admitting such a form is *symmetric*.

We use the standard theorem that every finite-dimensional semisimple $\mathbb{k}$ -algebra is symmetric. We do not prove this structure theorem here. The forms needed in our basic examples are explicit.

Example 1.30. For $S = \mathbb{k}^{\oplus r}$, take $\tau(a_1, \dots, a_r) = \sum_i a_i$. For $S = \text{Mat}_n(\mathbb{k})$, take $\tau(a) = \text{Tr}(a)$. Indeed, $\text{Tr}(ab) = \text{Tr}(ba)$ and $\text{Tr}(ae_{ji}) = a_{ij}$, so the pairing is non-degenerate in every characteristic. More generally, the sum of the matrix traces works on every product $\prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k})$ in Theorem 1.4.

Proposition 1.31. Choose a symmetrising form τ on S . For every finite-dimensional S -bimodule M , there are isomorphisms of S -bimodules

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \quad \beta_M : {}^*M \longrightarrow D(M), \quad g \longmapsto \tau \circ g. \quad (1.5)$$

There is also an isomorphism $\gamma_M : M^{*\text{en}} \rightarrow M^*$ given by

$$\gamma_M(F)(m) = \sum_j a_j \tau(b_j) \quad \text{where} \quad F(m) = \sum_j a_j \otimes b_j^{\text{op}}. \quad (1.6)$$

The maps $\alpha_M, \beta_M, \gamma_M$ commute with precomposition by bimodule homomorphisms. The maps α_M and β_M also apply to right and left S -modules, respectively, with the opposite side on the duals.

Proof For $\lambda \in D(M)$, non-degeneracy of τ determines $f(m), g(m) \in S$ uniquely by

$$\tau(f(m)s) = \lambda(ms), \quad \tau(sg(m)) = \lambda(sm) \quad (s \in S, m \in M). \quad (1.7)$$

Equation (1.7) gives $f(ms) = f(m)s$, $g(sm) = sg(m)$, and $\alpha_M(f) = \beta_M(g) = \lambda$. Hence α_M, β_M are bijective. For bimodule compatibility, use (1.4) and

$$\tau(sf(tm)) = \tau(f(tm)s) = \tau(f(tms)), \quad \tau(g(ms)t) = \tau(tg(ms)) = \tau(g(tms)).$$

The form $\tau_{\text{en}}(a \otimes b^{\text{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en} . The formula (1.6) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \quad \alpha_M(\gamma_M(F)) = \tau_{\text{en}} \circ F.$$

The argument (1.7), applied to S^{en} and τ_{en} , makes $F \mapsto \tau_{\text{en}} \circ F$ an isomorphism. Thus γ_M is an isomorphism. Composition with τ and τ_{en} commutes with precomposition, giving naturality. $\square$

The identifications (1.5) and (1.6) depend on τ . For instance, $\tau(a_1, \dots, a_r) = \sum_i c_i a_i$ is symmetrising on $\mathbb{k}^{\oplus r}$ for any $c_i \neq 0$. No semisimplicity assumption on S^{en} was used.

Definition 1.32. For a graded S -bimodule M , define its *graded S -duals* and *graded two-sided dual* by

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \quad ({}^*)M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}), \quad M^{(*\text{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\text{en}},$$

where each indicated summand has degree i . The first construction also applies to graded right S -modules, and the second to graded left S -modules.

Parentheses on the stars distinguish the graded constructions from the full S -duals. In the notation (1.1),

$$M^{(*)} = \underline{\text{Hom}}_S(M, S), \quad ({}^*)M = \underline{\text{Hom}}_{S^{\text{op}}}(M, S).$$

Corollary 1.33. For a locally finite graded S -bimodule M , a choice of symmetrising form gives natural isomorphisms of graded S -bimodules

$$M^{(*)} \cong \mathbb{D}(M) \cong ({}^*)M \cong M^{(*\text{en})}.$$

In particular, $\mathbb{D}(M) \cong M^{(*)}$ for a graded right S -module, and $\mathbb{D}(M) \cong ({}^*)M$ for a graded left S -module.

Proof Apply Proposition 1.31 to M_{-i} and take the direct sum over $i \in \mathbb{Z}$. Every comparison preserves degree because τ has degree 0. $\square$

For a graded algebra A with $S = A_0$, Corollary 1.33 transports the A -action (1.2) to the graded S -duals of a locally finite graded A -module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \quad ga = \beta_M^{-1}(\beta_M(g)a) \quad \text{for left } M.$$

Here $\alpha_M(f) = \tau \circ f$ and $\beta_M(g) = \tau \circ g$ are the restrictions of (1.5); $a \in A$, $f \in M^{(*)}$, and $g \in ({}^*)M$. The restrictions to S agree with Definition 1.26. These remain duals over S , not duals taking values in A .

Example 1.34. Let $S = \mathbb{k}^{\oplus r}$ with coordinate idempotents $e_1, \dots, e_r$ and the form $\tau(e_i) = 1$. For an S -bimodule M , the decompositions and the reversal of sides are

$$M = \bigoplus_{i,j} e_i M e_j, \quad e_j D(M) e_i \cong D(e_i M e_j) \cong e_j M^* e_i \cong e_j ({}^* M) e_i.$$

For the degree 1 piece $V = \mathbb{k}e_{12}$ of $\mathbb{k}A_2$, we have $e_1 V e_2 = V$. Its graded $\mathbb{k}$ -linear dual is spanned by ε of degree -1 , with $\varepsilon(e_{12}) = 1$ and $e_2 \varepsilon e_1 = \varepsilon$. The corresponding elements of V^* and *V send e_{12} to e_2 and e_1 , respectively. Thus equal scalar-valued functionals can correspond to different S -valued maps.

Lemma 1.35. For finite-dimensional right S -modules M and left S -modules N , evaluation gives canonical isomorphisms $M \cong {}^*(M^*)$ and $N \cong ({}^*N)^*$, sending m to $(f \mapsto f(m))$ and n to $(g \mapsto g(n))$, respectively. For bimodules, both are bimodule isomorphisms. No symmetrising form is needed.

Proof By Theorem 1.4, every finite-dimensional S -module is a direct summand of some $S^{\oplus n}$. Evaluation is an isomorphism for S , hence for such summands; (1.4) gives bimodule compatibility. $\square$

2 Lecture 2: Tensor and quadratic algebras

Tensor products over an algebra

The degree zero algebra acts on both sides of the degree 1 piece. We therefore need the tensor product over an algebra, not only the tensor product over $\mathbb{k}$.

Definition 2.1. Let M be a right S -module, let N be a left S -module, and let W be a $\mathbb{k}$ -vector space. The **tensor product** $M \otimes_S N$ is the quotient of $M \otimes_{\mathbb{k}} N$ by the subspace generated by

$$ms \otimes n - m \otimes sn \quad (m \in M, n \in N, s \in S).$$

Equivalently, a bilinear map $b : M \times N \rightarrow W$ factors uniquely through a linear map $M \otimes_S N \rightarrow W$ precisely when $b(ms, n) = b(m, sn)$ for all $m \in M, s \in S$, and $n \in N$.

The equality $ms \otimes_S n = m \otimes_S sn$ allows a scalar from S to pass between adjacent factors. For an S -bimodule V , define

$$V^{\otimes_S 0} := S, \quad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Tensor duality reverses the order of the factors.

Proposition 2.2. Let S be finite-dimensional semisimple and let V, W be finite-dimensional S -bimodules. There are natural bimodule isomorphisms

$$\begin{aligned} \theta_D : D(W) \otimes_S V^* &\xrightarrow{\sim} D(V \otimes_S W), \\ \theta_r : W^* \otimes_S V^* &\xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_\ell : {}^*W \otimes_S {}^*V &\xrightarrow{\sim} {}^*(V \otimes_S W), \\ \theta_{\text{en}} : W^* \otimes_S V^{*\text{en}} &\xrightarrow{\sim} (V \otimes_S W)^{*_{\text{en}}}, \end{aligned} \tag{2.1}$$

defined by

$$\begin{aligned} \theta_D(\lambda \otimes f)(v \otimes w) &= \lambda(f(v)w), \\ \theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \\ \theta_{\text{en}}(g \otimes F)(v \otimes w) &= \sum_j g(a_j w) \otimes b_j^{\text{op}}, \quad \text{where } F(v) = \sum_j a_j \otimes b_j^{\text{op}}. \end{aligned} \tag{2.2}$$

Proof The formulas in (2.2) respect the tensor relations and bimodule actions. The maps θ_D and θ_r are isomorphisms for $V = S$ as a right module, hence for finite direct sums and direct summands of S . For θ_ℓ , apply the same argument to the left module W . Choose a symmetrising form τ as in Definition 1.29, and use $\gamma_V, \gamma_{V \otimes_S W}$ from (1.6). Then

$$\gamma_{V \otimes_S W} \circ \theta_{\text{en}} = \theta_r \circ (\text{id}_{W^*} \otimes \gamma_V),$$

since both sides evaluate $g \otimes F$ at $v \otimes w$ as $\sum_j g(a_j w) \tau(b_j)$. Proposition 1.31 makes both γ maps isomorphisms, so θ_{en} is an isomorphism. $\square$

For a fixed symmetrising form τ , let $\alpha_M(f) = \tau \circ f$ as in (1.5). Then tensor evaluation satisfies

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \text{id}_{V^*}).$$

Both sides send $g \otimes f$ to $v \otimes w \mapsto \tau(g(f(v)w))$.

Example 2.3. Let $S = \mathbb{k}^{\oplus r}$ with coordinate idempotents $e_1, \dots, e_r$, and suppose $V = e_i V e_j$ and $W = e_j W e_\ell$. By (1.4),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \quad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus θ_r in (2.1) reverses the two outer idempotents.

Generated algebras

Let A be graded with $S = A_0$. Multiplication makes A_1 an S -bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \quad a_1 \otimes \dots \otimes a_i \longmapsto a_1 \dots a_i \quad (i \geq 1), \quad \mu_0 = \text{id}_S. \quad (2.3)$$

Definition 2.4. The algebra A is *generated in degree 1 over S* if every μ_i in (2.3) is surjective.

Equivalently, S and A_1 generate A as an algebra.

Example 2.5. The algebra $\mathbb{k}A_n$ of Example 1.7 is generated in degree 1 over $S = \bigoplus_i \mathbb{k}e_{ii}$:

$$e_{ij} = e_{i,i+1}e_{i+1,i+2} \dots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading $\deg e_{ij} = 2(j-i)$, the same algebra has $A_1 = 0$ and $A_2 \neq 0$, so it is not generated in degree 1.

Tensor algebras and the universal property

Definition 2.6. Let S be a $\mathbb{k}$ -algebra and V an S -bimodule. The *tensor algebra of V over S* is

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given S -actions.

The left and right S -actions determine which factors can be concatenated. For $S = \bigoplus_{i=1}^r \mathbb{k}e_i$ with coordinate idempotents,

$$V = \bigoplus_{i,j} e_i V e_j, \quad (e_i V e_j) \otimes_S (e_\ell V e_m) = 0 \quad \text{if } j \neq \ell. \quad (2.4)$$

Indeed, $v \otimes w = v e_j \otimes w = v \otimes e_j w = 0$ when $v \in e_i V e_j$, $w \in e_\ell V e_m$, and $j \neq \ell$.

Example 2.7. Let $S = \mathbb{k}^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3 , and let $V = \mathbb{k}a \oplus \mathbb{k}b$ have actions $e_1 a e_2 = a$ and $e_2 b e_3 = b$, with all other idempotent actions zero. By (2.4),

$$V \otimes_S V = \mathbb{k}(a \otimes b), \quad V^{\otimes_S 3} = 0, \quad T_S(V) = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ab.$$

Here $ab = a \otimes b$. The assignments $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, and $b \mapsto e_{23}$ give $T_S(V) \cong \mathbb{k}A_3$; the tensor ab maps to e_{13} .

Proposition 2.8 (Universal property of the tensor algebra). Let $\iota_S : S \rightarrow T_S(V)$ and $\iota_V : V \rightarrow T_S(V)$ be the canonical inclusions. Given a unital $\mathbb{k}$ -algebra homomorphism $\varphi : S \rightarrow B$ and an S -bimodule map $f : V \rightarrow B$, where $sbt = \varphi(s)b\varphi(t)$ on B , there is a unique $\mathbb{k}$ -algebra homomorphism $\tilde{f} : T_S(V) \rightarrow B$ making the diagram commute:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow f & \downarrow \exists! \tilde{f} & \swarrow \varphi & \\ & & B & & \end{array} \quad (2.5)$$

Proof Define $\tilde{f}$ linearly by

$$\tilde{f}(s) = \varphi(s), \quad \tilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1) \cdots f(v_i) \quad (s \in S, i \geq 1). \quad (2.6)$$

The identity $f(svt) = \varphi(s)f(v)\varphi(t)$ makes $\tilde{f}$ well-defined on tensor products and compatible with multiplication. Every homomorphism satisfying (2.5) has the values in (2.6), proving uniqueness. $\square$

Corollary 2.9. *If A is generated in degree 1 over $S = A_0$, there is a surjective graded algebra homomorphism $\mu : T_S(A_1) \twoheadrightarrow A$ whose degree i component is μ_i from (2.3).*

Proof Apply Proposition 2.8 to the inclusions $S \rightarrow A$ and $A_1 \rightarrow A$. Surjectivity of each μ_i gives surjectivity of μ . $\square$

Definition 2.10. *For the map μ in Corollary 2.9, the homogeneous two-sided ideal $I := \text{Ker } \mu$ is the ideal of relations.*

Writing $V = A_1$, we obtain $A \cong T_S(V)/I$ with $I_0 = I_1 = 0$. The bimodule V specifies the generators and their left and right S -actions; I specifies the relations between their products.

Quadratic algebras

Definition 2.11. *Let S be finite-dimensional semisimple, V a finite-dimensional S -bimodule, and $R \subset V \otimes_S V$ an S -sub-bimodule. The algebra $T_S(V)/(R)$ is a quadratic algebra, where (R) denotes the two-sided ideal generated by R .*

Proposition 2.12. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$ and $A_1 = V$, is locally finite, and is generated in degree 1 over S .*

Proof The ideal (R) has no components in degrees 0 and 1. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and multiplication maps $V^{\otimes_S i}$ onto A_i . $\square$

Example 2.13. *Let $S = \bigoplus_{i=1}^n \mathbb{k}e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$ inside $\mathbb{k}A_n$. By (2.4), the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \mapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Thus multiplication gives $T_S(V) \cong \mathbb{k}A_n$, a quadratic algebra with $R = 0$.

Example 2.14. *For S, V, a, b from Example 2.7, the quotient*

$$T_S(V)/(ab) \cong \mathbb{k}A_3/(e_{13})$$

is quadratic with $R = \mathbb{k}(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b , in degrees 0, 0, 0, 1, 1, as in Example 1.19.

Example 2.15. *Let $S = \mathbb{k}^{\oplus 2}$ with coordinate idempotents e_1, e_2 , and let $V = \mathbb{k}a \oplus \mathbb{k}b$ satisfy $e_1 a e_2 = a$ and $e_2 b e_1 = b$, with all other idempotent actions zero. In $T_S(V)$ only alternating words in a, b survive. The quadratic algebra $A = T_S(V)/(ab)$ has*

$$A = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ba, \quad \dim_{\mathbb{k}} A_0 = 2, \quad \dim_{\mathbb{k}} A_1 = 2, \quad \dim_{\mathbb{k}} A_2 = 1.$$

Every alternating word of length at least 3 contains ab , so all higher components vanish.

Example 2.16. *Let $B = \mathbb{k}A_4$ and $A = B/(e_{14})$, with the grading of Example 1.7. The ideal $I = (e_{14}) = \mathbb{k}e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12}e_{23}e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. Hence A is not quadratic.*

Example 2.17 (Polynomial and exterior algebras). For the classical comparison, take $S = \mathbb{k}$ and $V = \bigoplus_{i=1}^n \mathbb{k}x_i$. The [symmetric algebra](#) and [exterior algebra](#) are

$$\begin{aligned} \text{Sym}(V) &= T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \mathbb{k}[x_1, \dots, x_n], \\ \bigwedge V &= T_{\mathbb{k}}(V)/(v^2 \mid v \in V). \end{aligned} \tag{2.7}$$

Both are quadratic. The polynomial basis allows repetitions; the exterior basis consists of 1 and $x_{i_1} \cdots x_{i_d}$ with $i_1 < \cdots < i_d$. The exterior relations are equivalently $x_i^2 = 0$ for $1 \leq i \leq n$ and $x_i x_j + x_j x_i = 0$ for $i < j$, in every characteristic.

Hilbert series

A Hilbert series records the dimension of each homogeneous component.

Definition 2.18. Let $M = \bigoplus_{j \in \mathbb{Z}} M_j$ be a graded vector space with $\dim_{\mathbb{k}} M_j < \infty$ for all j and $M_j = 0$ for all $j \ll 0$. Its [Hilbert series](#) is the formal Laurent series $h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j$.

For a locally finite graded algebra A , $h_A(t)$ refers to its underlying graded vector space; the series does not record multiplication.

Lemma 2.19. Let M, N be locally finite graded vector spaces concentrated in degrees bounded below.

- (1) A graded isomorphism $M \cong N$ implies $h_M(t) = h_N(t)$.
- (2) An exact sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ with degree 0 maps gives $h_M(t) = h_L(t) + h_N(t)$.
- (3) For every $d \in \mathbb{Z}$, $h_{M(d)}(t) = t^{-d} h_M(t)$.

Proof The first two statements follow from dimensions in each degree. For shifts,

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i. \quad \square$$

Example 2.20. For the upper triangular algebra in Example 1.7, counting matrix entries on each diagonal gives

$$h_{\mathbb{k}A_n}(t) = n + (n-1)t + \cdots + t^{n-1}, \quad h_{\mathbb{k}A_3/(e_{13})}(t) = 3 + 2t.$$

The second equality also follows by subtracting $h_{(e_{13})}(t) = t^2$ from $h_{\mathbb{k}A_3}(t)$, as in Lemma 2.19(2). For $A = T_S(V)/(ab)$ in Example 2.15, the basis e_1, e_2, a, b, ba gives $h_A(t) = 2 + 2t + t^2$.

Example 2.21. For the algebras in Example 2.17, each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus

$$h_{\text{Sym}(V)}(t) = \frac{1}{(1-t)^n}, \quad h_{\bigwedge V}(t) = (1+t)^n \quad \text{where } n = \dim_{\mathbb{k}} V.$$

Lemma 2.22. For locally finite graded vector spaces V, W concentrated in degrees bounded below,

$$h_{V \otimes_{\mathbb{k}} W}(t) = h_V(t) h_W(t), \quad (V \otimes_{\mathbb{k}} W)_n = \bigoplus_{i+j=n} V_i \otimes_{\mathbb{k}} W_j.$$

Proof The coefficient of t^n on either side is $\sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j$. Only finitely many summands are non-zero because both gradings are bounded below. $\square$

The degree 2 relations of a quotient

Let S be finite-dimensional semisimple and V a finite-dimensional S -bimodule. Let $A = T_S(V)/I$ with $I_0 = I_1 = 0$ as in Definition 2.10. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where} \quad I_2 = I \cap (V \otimes_S V).$$

Definition 2.23. The *quadratic part* of A is $qA := T_S(V)/(I_2)$.

The inclusion $(I_2) \subset I$ induces a surjective graded homomorphism $\pi_A : qA \twoheadrightarrow A$, and

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

Example 2.24. For $A = \mathbb{k}A_4/(e_{14})$ from Example 2.16, we have $I_2 = 0$, so

$$qA = \mathbb{k}A_4, \quad \pi_A : \mathbb{k}A_4 \twoheadrightarrow \mathbb{k}A_4/(e_{14}), \quad \text{Ker } \pi_A = \mathbb{k}e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

Orthogonal complements and quadratic duality

Definition 2.25. For an S -sub-bimodule U of a finite-dimensional S -bimodule M , define the *right orthogonal complement* and *left orthogonal complement* by

$$U^\perp := \{f \in M^* \mid f(U) = 0\}, \quad {}^\perp U := \{g \in {}^*M \mid g(U) = 0\}.$$

Both $U^\perp$ and ${}^\perp U$ are S -sub-bimodules by (1.4). Fix a symmetrising form τ as in Definition 1.29. For the comparison maps $\alpha_M(f) = \tau \circ f$ and $\beta_M(g) = \tau \circ g$ in (1.5),

$$\alpha_M(U^\perp) = \beta_M({}^\perp U) = \{\lambda \in D(M) \mid \lambda(U) = 0\}. \quad (2.8)$$

Indeed, if $\alpha_M(f)(U) = 0$, then

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S, u \in U,$$

so non-degeneracy of τ gives $f(U) = 0$. For β_M , use $\tau(sg(u)) = \tau(g(su))$ instead.

For $A = T_S(V)/(R)$, the map θ_r in (2.1) identifies $R^\perp$ with a sub-bimodule of $V^* \otimes_S V^*$. These are the generators and relations used for the quadratic dual in Lecture 3:

$$V^* = \text{Hom}_S(V, S), \quad R^\perp \subset V^* \otimes_S V^*.$$

The generators V^* are assigned degree 1. In contrast, the graded S -dual $V^{(*)}$ from Definition 1.32 has degree -1 when $V = A_1$.

Example 2.26. For $\text{Sym}(V)$ and $\bigwedge V$ in Example 2.17, take $V = \mathbb{k}x \oplus \mathbb{k}y$. The relation spaces are

$$\begin{aligned} R_{\text{Sym}(V)} &= \mathbb{k}(x \otimes y - y \otimes x), \\ R_{\bigwedge V} &= \mathbb{k}(x \otimes x) \oplus \mathbb{k}(y \otimes y) \oplus \mathbb{k}(x \otimes y + y \otimes x). \end{aligned}$$

Let $x^*, y^* \in V^*$ be the coordinate functionals. With the tensor evaluation θ_r in (2.2),

$$\begin{aligned} R_{\text{Sym}(V)}^\perp &= \text{span}_{\mathbb{k}}\{x^* \otimes x^*, y^* \otimes y^*, x^* \otimes y^* + y^* \otimes x^*\}, \\ R_{\bigwedge V}^\perp &= \mathbb{k}(x^* \otimes y^* - y^* \otimes x^*). \end{aligned}$$

Thus the quadratic dual presentations exchange the symmetric and exterior relations.

Exercise 2.27. Let $A = \mathbb{k}A_4/(e_{13}, e_{24})$ with the grading of Example 1.7. Find a homogeneous basis, compute $h_A(t)$, and write $A = T_S(V)/(R)$ with explicit S, V, R .

Exercise 2.28. For the bimodule V in Example 2.15, find a basis of each $T_S(V)_d$ and show that $h_{T_S(V)}(t) = 2/(1-t)$. Compare with the quotient $T_S(V)/(ab)$.

Exercise 2.29. For the algebras in Example 2.17, prove

$$\dim_{\mathbb{k}} \operatorname{Sym}(V)_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{k}} \bigwedge^2 V = \frac{n(n-1)}{2} \quad \text{where } n = \dim_{\mathbb{k}} V.$$

References

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